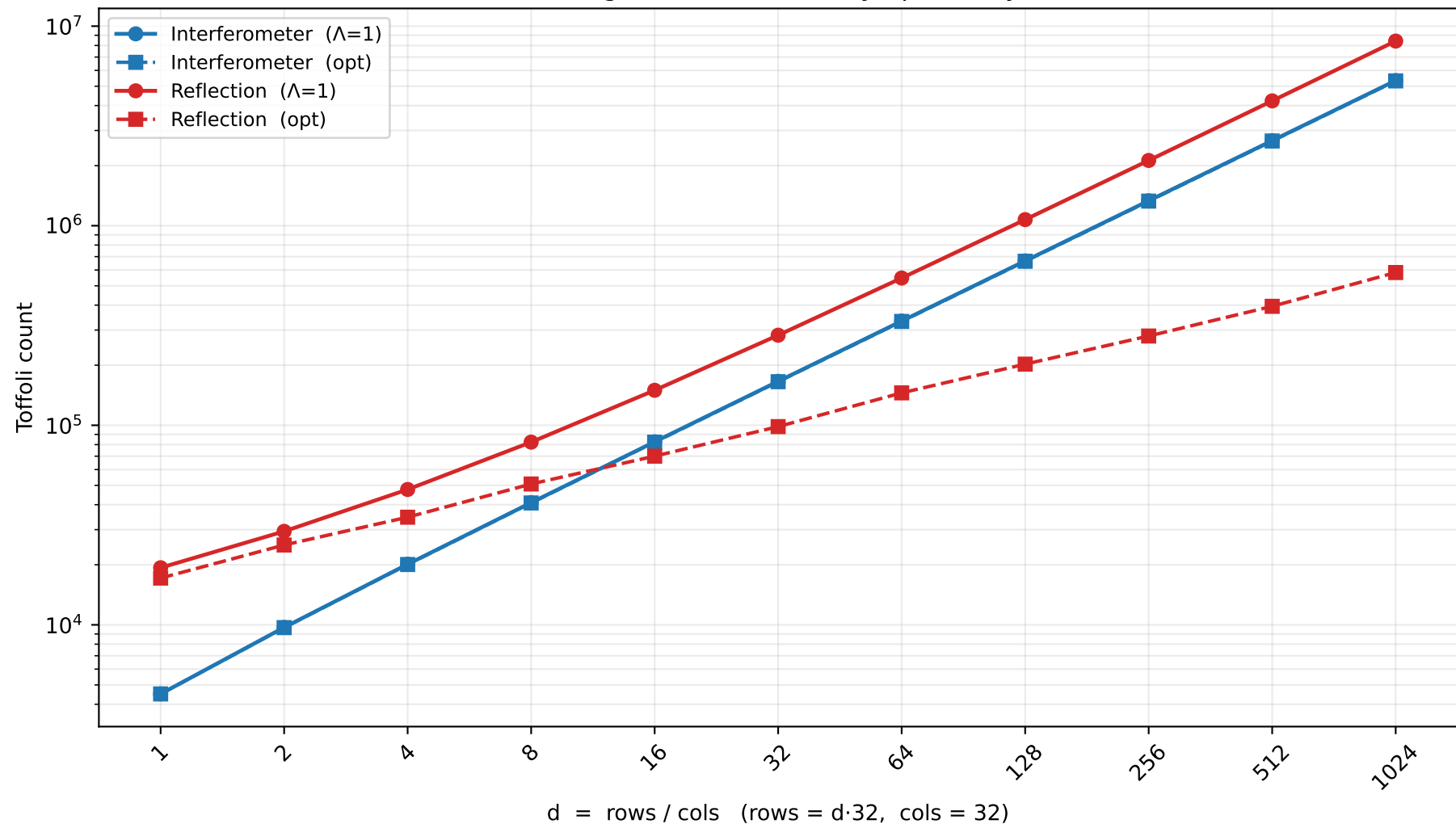
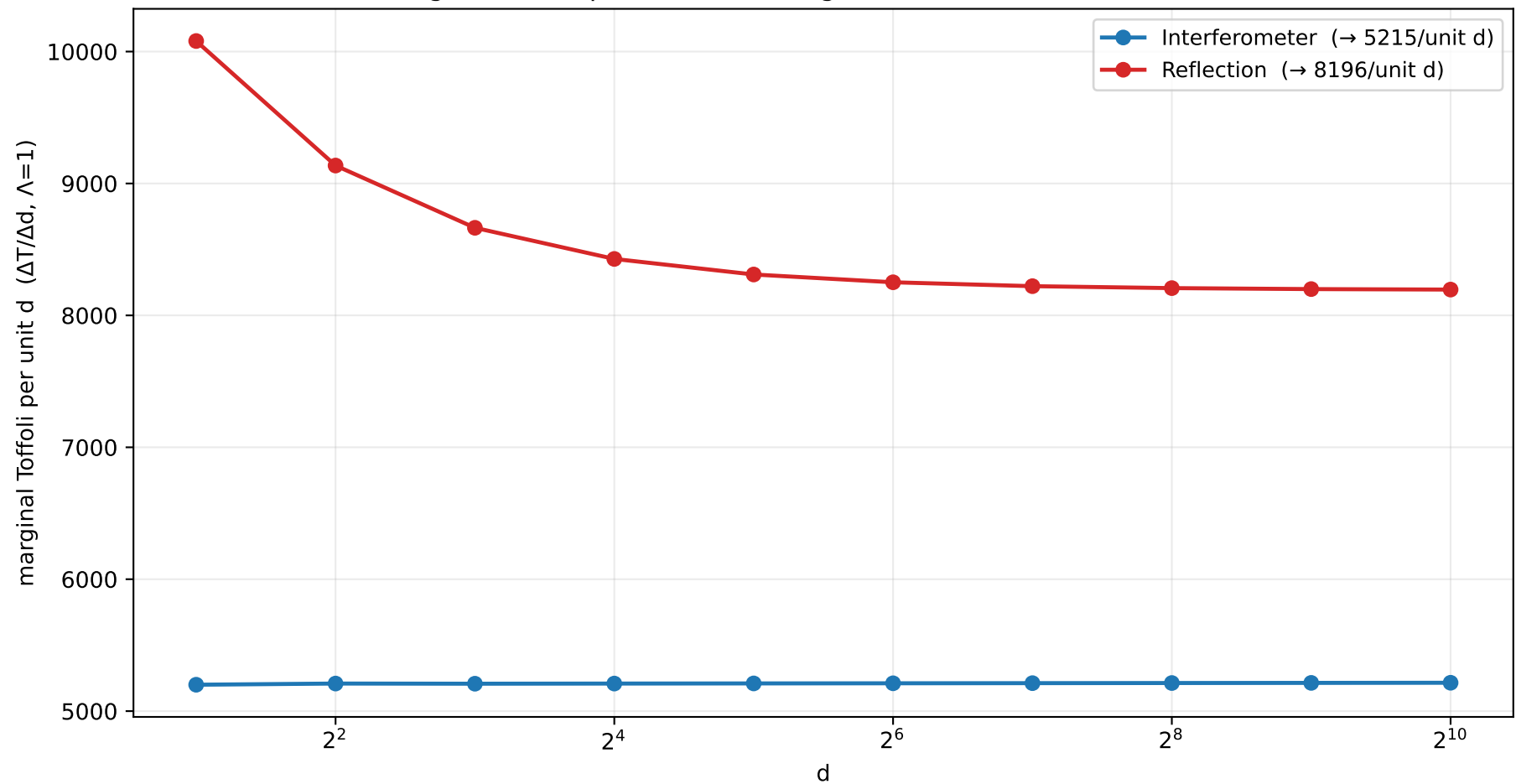


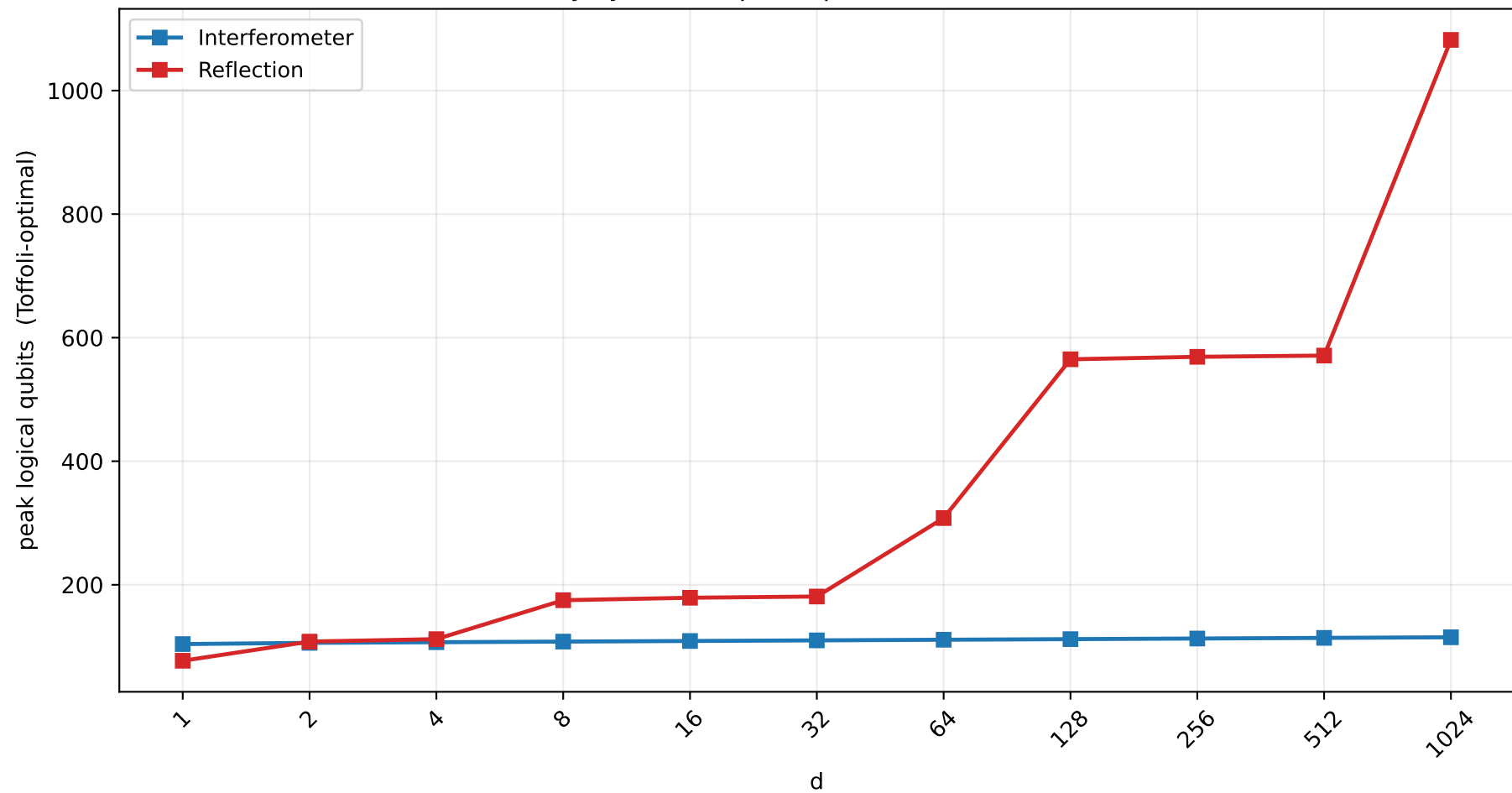
Isometry synthesis ($d \cdot N \times N$): interferometer vs reflection, Toffoli vs d
($N=32$, $b=32$, single block) -- both asymptotically LINEAR in d



Marginal Toffoli per unit d converges to a constant $\Rightarrow T$ linear in d



Isometry synthesis: peak qubits vs d ($N=32$, $b=32$)



Isometry synthesis Toffoli: interferometer vs reflection (N=32, b=32)
large-d fit (d≥64): Interferometer: $\Lambda=1$ d^{1.00}, opt d^{1.00} Reflection: $\Lambda=1$ d^{0.99}, opt d^{0.50}

d	rows	Interferometer T($\Lambda=1$)	Reflection T($\Lambda=1$)	Interferometer T(opt)	Reflection T(opt)
1	32	4,505	19,328	4,505	17,152
2	64	9,705	29,408	9,684	25,120
4	128	20,123	47,680	20,060	34,624
8	256	40,954	82,336	40,807	50,784
16	512	82,625	149,760	82,310	70,016
32	1024	165,984	282,720	165,333	98,464
64	2048	332,735	546,752	331,412	145,344
128	4096	666,302	1,072,928	663,635	202,464
256	8192	1,333,565	2,123,392	1,328,210	280,064
512	16384	2,668,348	4,222,432	2,657,617	394,528
1024	32768	5,338,427	8,418,624	5,316,944	582,720